



Research on the prediction and influencing factors of heavy duty truck fuel consumption based on LightGBM

Xinyi Zhu^a, Xiaoyan Shen^{a,*}, Kailiang Chen^b, Zeqing Zhang^a

^a School of Automobile, Chang'an University, Xi'an, 710016, China

^b BYD Co Ltd, Xi'an, 710016, China

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ABSTRACT

Excessive fuel consumption of heavy duty trucks aggravates environmental pollution and leads to reduced economic benefits of road transportation enterprises. In order to study the influencing factors of fuel consumption of heavy duty trucks, based on real-time vehicle operating parameter data, this paper establishes a segment fuel consumption prediction model based on LightGBM algorithm. The fitting ability of the model is improved by feature selection and Optuna hyperparameter search. The performance is compared with other models. The results show that the prediction effect of LightGBM model is optimal. Finally, based on the SHAP framework, the prediction results of the optimal fragmented fuel consumption model are visually explained. The conclusion that average speed and average throttle pedal value have a significant impact on segmented fuel consumption. Ensuring higher and more stable speeds, reducing the frequency of acceleration and deceleration can reduce the fuel consumption appropriately.

1. Introduction

Road transport is the dominant mode of transportation and key to global economic development. As the economy develops, road mileage rises year after year and road transport takes a greater share in the transportation sector [1]. As the main carrier for undertaking road transportation tasks, heavy duty trucks play an important role in the development of the transportation industry. Due to its long body, heavy load and long running time, it causes a lot of fuel consumption and environmental pollution, which has a negative impact on economic efficiency [2]. According to statistics, the transportation industry in China consumes more than 50% of the total amount of diesel fuel annually [3], and most of it comes from heavy duty trucks. At the same time, heavy duty trucks have become a major source of emission pollution. In 2021, automobile NO_x and PM emissions in China will be 5.685 million tons and 65,000 tons respectively. Among them, the emissions from heavy duty trucks account for 76.1% and 91.3% of the total, bringing a huge burden to people's health [4]. At present, China uses WHTC (World Harmonized Transient Cycle) and WHSC (World Harmonized Steady-state Cycle) to test fuel consumption and emissions when producing heavy duty truck engines according to the National VI standard [5]. However, in the actual process, the heavy truck transportation section is concentrated in the country roads and highways, urban road operating conditions accounted for less. Subject

to serious traffic congestion constraints, the vehicle exists more frequent acceleration and deceleration transitions and idling in neutral. These factors cause the actual fuel consumption to be different from the test values under the above test conditions [6,7]. Therefore, it is of great theoretical and practical significance to establish an accurate fuel consumption prediction model based on the actual working condition data of heavy trucks, analyze the influencing factors of fuel consumption, and take measures to reduce fuel consumption [8].

There is a close relationship between driving behavior and fuel consumption. Some scholars analyzed the degree of influence of different driving behaviors on fuel consumption by conducting statistics. Zorrofi [9] presented a model for a series hybrid bus system, concluding that erratic driving patterns lead to reduced fuel efficiency. Carres [10] used speed standard deviation, number of stops, and acceleration phase as driving behavior evaluation indexes to analyze the driving data of buses. The results showed that reasonable driving behavior can save 27% of fuel. Thomas [11] found that a steady driving style can reduce fuel consumption of hybrid vehicles by 25% to 68% from the original vehicle. Quantitative analysis for specific driving behavior indicators is a more specific direction to study fuel consumption. Lee [12] collected behavioral parameters and corresponding fuel consumption of 15 young drivers driving on the highway. It was concluded that the average depth of the accelerator pedal had a significant effect on fuel

* Corresponding author.

E-mail address: sxy719@163.com (X. Shen).

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efficiency during highway driving in the highest gear. Jamson [13] investigated the prioritization of fuel economy among three existing eco-driving strategies. The aim was to enable the vehicle to achieve optimal fuel efficiency by providing advice on gas pedal angle. Zhang [14] analyzed socio-demographic characteristics and driving behavior in driving data from highway intersections. The results showed that motor vehicles traveling at higher speeds have better fuel efficiency. Thitipatanapong [15] explored the relationship between driver behavior and fuel consumption by collecting vehicle data. The results showed that fuel consumption was significantly correlated with acceleration rate. Lois [16] analyzed the factors affecting fuel consumption and ecological driving by conducting an ecological driving experiment on Spain. The results showed that eco-driving is strongly influenced by driving behaviors such as deceleration rate, RPM and speed.

The establishment of an accurate vehicle fuel consumption model can provide a theoretical basis for energy conservation and emission reduction. The typical macro models, the MOBILE model [17] and the COPERT model [18], have been widely used internationally. These models project the total fuel consumption and emissions of transportation in a city or country based on the average speed of traffic flow in a certain area. The microscopic model is mainly derived based on the relationship between vehicle fuel consumption and vehicle transient state parameters. Walnum [19] used multivariate regression analysis and corresponding mean elasticity analysis to regress predictions on data generated by an online fleet management system for a Norwegian heavy duty trucking company. Hoffman [20] focused on the effect of road inclination and payload on fuel consumption. Multiple linear regression models were developed to predict fuel consumption for different routes. With the development of machine learning techniques, Slavin [21] took the body parameters – torque, load capacity, wheel-base – as inputs and body fuel consumption as output. Modeling each using linear regression and neural network approaches yielded better performance for neural network predictions. Moradi [22] proposed an artificial neural network (ANN) modeling approach. The engine speed was introduced into the model through a cascade modeling procedure, and the results showed that the improved model has a higher accuracy. Wickramanayake [23] compared the prediction ability of three machine learning methods in terms of fuel consumption of buses, and the results showed that the Random Forest prediction model is more accurate. Bousonville [24] established the fuel consumption prediction models in conjunction with weather conditions. It indicated that the Gradient Boosting regression model showed better prediction results compared to the traditional machine learning K-nearest Neighbor regression model and Artificial Neural Network regression model. Topi [25] built a neural network-based fuel consumption model for buses and found that using road gradient as an input parameter can effectively improve the prediction accuracy of the model.

Comparing the results of existing studies, from the traditional multiple linear regression methods to the current development of machine learning techniques, the research on fuel consumption has been continuously deepened. Algorithms such as tree modeling, integrated learning, and even neural network modeling are widely used in fuel consumption prediction, and certain results have been achieved. For heavy duty trucks, which consume more fuel, most of the existing research has focused on Random Forest and plain Bayesian algorithms. Due to the lack of interpretability of machine learning models, most of the research on fuel consumption based on machine learning has only shown the performance of the trained models, or relied on statistics to simply characterize the data features. There is no in-depth analysis of the influencing factors, which cannot explain the influence of each feature on fuel consumption. Therefore, this paper is based on the Lightweight Gradient Boosting Machine (LightGBM) algorithm to improve the prediction accuracy of fuel consumption of heavy duty trucks. The SHAP framework is selected as a model interpretation tool to analyze the influence of driving behavior parameters on fuel consumption. This provides a theoretical basis for regulating driving behavior and achieving emission reductions.

2. Data

2.1. Data acquisition

The data samples in this paper are collected from three heavy duty trucks between June 14 and October 30, 2022 by a logistics company in China. In order to ensure the reliability of the data, high-precision OBD data acquisition equipment [26] is used, and strict data quality control is carried out. To ensure that each vehicle completes its transportation tasks on a daily basis, a total of 138 days of instantaneous operating data are collected on conventional roads, covering key indicators such as vehicle speed, rotational speed, engine load, and so on. Raw data volume of over 3 million. The field list of vehicle operating condition information collected by OBD is shown in Table 1.

2.2. Data preprocessing

In the data acquisition and transmission phase, the original data may be interfered by various factors. Signal interruption, bad weather, and abnormal vehicle movement can lead to problems such as duplicate recording, partially missing data, and abnormal data. In order to ensure that the data are stable and reliable, this study will take effective measures to deal with these problems.

2.2.1. Missing data

Missing data generally exists in two situations: a large number of continuous missing and a small number of sporadic missing. For more than 5 rows of continuous missing data in a column, these rows of data are directly deleted to avoid the result of inaccuracy. For sporadic missing data, prediction filling on the basis of the time series is a more appropriate method. As shown in Eq. (1), due to the high frequency of data collection in this paper and the close relationship between adjacent data, using the simple averaging method can achieve good filling effect with less computational complexity:

$$F_n = \frac{1}{k}(F_{n-1} + F_{n-2} + \dots + F_{n-k}) \quad (1)$$

where F_n is the missing data, F_{n-i} is the first 1 s of data, k is the number of reference hours selected, typically 10.

2.2.2. Duplicate data

The data collected through OBD is time-series data. For each vehicle, only one valid piece of data is generated per second. Data other than time may be duplicated in some cases, especially such as fuel tank level, engine coolant temperature, and atmospheric pressure. Completely duplicated data reduces the diversity and information content of the dataset, which affects the accuracy and reliability of analysis. In order to improve the analytical performance of the model, duplicate values need to be identified and removed. For these duplicate data samples, the method of retaining the first row of data and removing other duplicates is used for processing.

2.2.3. Abnormal data

Abnormal data refers to data points with unusual values in a dataset. Their significant differences from other data points can have a negative impact on the experimental results. To address this issue, techniques such as interpolation can be used to fill in missing data. Further cleaning and verification of other data to ensure data integrity and accuracy.

The original working condition data has a large scale and can be efficiently batch processed with the help of Pandas, a data processing library in Python.

2.3. Set kinematic segments

The kinematic segment is the process from ignition start to ignition off stop of the vehicle. It can characterize the driving behavior and resulting fuel consumption in this state. The preprocessed data will be divided into kinematic segments.

Table 1
Data information collection fields table.

Data collection field name	Unit	Data collection field name	Unit	Data collection field name	Unit
Device Id	–	NOx upstream of SCR	ppm	Longitudes	°
Trip Id	–	NOx downstream of SCR	ppm	Latitude	°
Measuring time	s	Reactant Residual	%	Accumulated mileage	km
Vehicle speed	km/h	Air inflow	kg/h	Accelerator pedal	%
Atmospheric pressure	kPa	SCR inlet temperature	°C	Accumulated fuel consumption	L
Net engine output torque	%	SCR outlet temperature	°C	Urea tank temperature	°C
Friction torque	%	DPF differential pressure	kPa	Actual urea injection volume	ml/h
Engine revolutions	rpm	Engine coolant temperature	°C	DPF exhaust temperature	°C
Engine fuel flow	L/h	Fuel tank level	%	Milestone	m

Table 2
Number of kinematic segments at different time length thresholds.

Time length thresholds (s)	Number of clips	Number of raw data included
1800	728	2578611
1200	961	2924320
600	1153	3076253

2.3.1. Fuel economy evaluation indicators

There are two commonly used units for evaluating fuel economy: L/h (or mL/s) and L/100 km. The former is more concerned with fuel consumption in the instantaneous state or consumption per hour at a constant speed. The latter is more suitable for evaluating and characterizing the fuel consumption of vehicles under certain operating conditions. Considering that this paper explores the effect of driving behavior on fuel consumption of kinematic segments, the raw fuel consumption data is transformed.

As shown in Eq. (2), calculate the total fuel consumption of the segment:

$$V_{total} = \frac{\sum_1^n V_{fuel}}{3600} \quad (2)$$

where V_{fuel} is the instantaneous fuel consumption taken per second in L/h. The total fuel consumption value V_{total} , in L, for this period was found by combining the calculations.

Multiply the ratio of 100 km to the mileage of the segment and the fuel consumption value of that segment. As shown in Eq. (3), the fuel consumption per 100 kilometers for this segment is generated.

$$Q = V_{total} \times \frac{100 \times 10^3}{S} \quad (3)$$

where S is the distance traveled by the segment in m.

As shown in Eq. (4), the final formula for calculating fuel consumption is obtained.

$$Q = \frac{100 \times \sum_1^n V_{fuel}}{3.6 \times S} \quad (4)$$

2.3.2. Time threshold

When some vehicles are idling for a long time, the engine still consumes fuel, resulting in the fuel consumption per 100 kilometers calculated by the above method is much greater than that of other vehicles. To eliminate the influence of long idling time on the results, this paper defines the kinematic segment as the actual driving process in which the time difference between ignition and flameout of a vehicle is greater than a certain threshold and the average speed is greater than 20 km/h. In order to have a certain scale of the number of kinematic segments, the time thresholds are set to generate different numbers of segments. As in Table 2, the time thresholds are set to 1800 s, 1200 s, and 600 s, respectively.

The number of kinematic segments obtained increases as the kinematic segment duration threshold decreases. Correspondingly, the amount of original data that can be utilized increases as well. Therefore, the time length threshold is set to 600 s in this paper. The segment duration and the total mileage are used as the descriptive features of the kinematic segments.

3. Methods

3.1. Fundamentals

The LightGBM [27] algorithm, developed by Microsoft, is an upgraded version of Gradient Boosting Decision Tree (GBDT) and XGBoost [28]. XGBoost uses pre-sorted algorithm which is more space and time consuming. The indexing of feature values and their corresponding sample statistics must be recorded. The principle of LightGBM is multiple rounds of iteration. A weak classifier (CART decision tree) is generated by each round of iteration. Each classifier is trained on the residuals from the previous round of classifiers to improve prediction accuracy. It occupies less memory and consumes less time for computation.

According to the principle of LightGBM, let the dataset have n samples $S = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$. Where x_i is the feature and y_i is the corresponding label. The prediction result is shown in Eq. (5):

$$f(x) = \sum_{i=1}^n w_{(x)} \quad (5)$$

where $f(x)$ is the sum of the output values ($w_{(x)}$) stacked training for each decision tree.

LightGBM adopts the Gradient-based One-Side Sampling (GOSS) technique. As shown in Eq. (6), it selects the samples with the larger gradient as subset A and the samples with the smaller gradient as subset B. The samples are partitioned by the variance gain $V_{(d)}$ of $A \cup B$:

$$V_j(d) = \frac{1}{n} \left[\frac{(\sum_{x_i \in A_l} g_i + \sum_{x_i \in B_l} g_i)^2}{n_l^j(d)} + \frac{(\sum_{x_i \in A_r} g_i + 1 - \frac{a}{b} \sum_{x_i \in B_r} g_i)^2}{n_r^j(d)} \right] \quad (6)$$

where $A_l = \{x \in A : x_{ij} < d\}$, $A_r = \{x \in A : x_{ij} > d\}$, $B_l = \{x \in A : x_{ij} < d\}$, $B_r = \{x \in A : x_{ij} > d\}$, g_i is the negative gradient of the LightGBM loss function at each iteration.

In addition to using this sampling technique, the LightGBM algorithm utilizes the EFB (Exclusive Feature Bundling) technique to optimize the training process. In the EFB technique, mutually exclusive features are composed into a set of feature histograms, thereby accelerating the feature processing process. This method shows strong robustness when dealing with complex situations such as Multi-collinearity. In summary, the LightGBM algorithm is based on two main features, GOSS and EFB, and has shown good performance in the field of integrated learning.

3.2. Feature selection

To describe and evaluate a kinematic fragment, the corresponding kinematic characteristic parameters should be selected. These kinematic parameters should be as comprehensive as possible to describe the kinematic segment according to its operating state and to form a mathematical model based on the kinematic segment. According to the

Table 3
Feature importance sorting based on LightGBM.

Feature name	Feature importance	Feature name	Feature importance
Average speed	0.25235	Maximum deceleration	0.031911
Average throttle pedal value	0.118652	Deceleration ratio	0.024788
Negative acceleration standard deviation	0.08033	Maximum speed	0.016564
Average positive acceleration	0.076006	Uniform speed ratio	0.016404
Rapid acceleration ratio	0.072185	Rapid deceleration ratio	0.01382
Average negative acceleration	0.069734	Overspeed ratio	0.01375
Throttle pedal standard deviation	0.05275	Maximum acceleration	0.013212
Long-duration idle ratio	0.047384	Positive acceleration ratio	0.011318
Positive acceleration standard deviation	0.039601	Speed standard deviation	0.007666
Total mileage	0.034435	Total time spent	0.007141

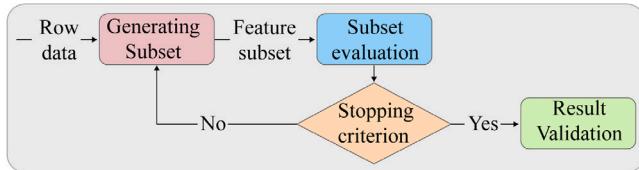


Fig. 1. Feature selection process based on LightGBM.

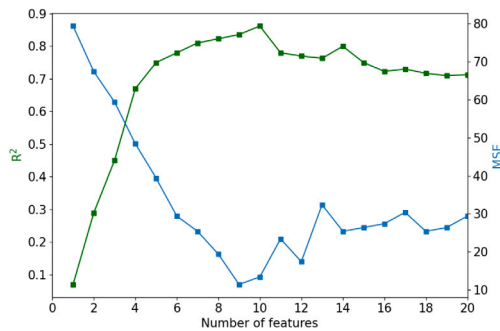


Fig. 2. Effect of number of features on model performance.

findings in related research [29,30], there are many factors affecting driving style and fuel consumption.

This paper initially selects 20 driving behavior indicators to describe the kinematic segment, including speed and acceleration derivatives, throttle pedal control degree, and other types of factors. It is ensured that the actual driving indicators of each segment can be fully characterized, and their relationship with fuel consumption can be deeply explored. Due to the large number of features in the data, relying solely on the Pearson correlation coefficient for feature selection is ineffective. Therefore, the features are selected with the help of the LightGBM algorithm, and the process framework is shown in Fig. 1.

First, the data are randomly divided into 80% test set and 20% training set, and all 20 features are input for model training using LightGBM algorithm. The obtained feature importance ranking is shown in Table 3.

Secondly, establish a list containing all features, and input 20 features in order of importance from this list. The segmented fuel consumption regression model is constructed using LightGBM. The optimal number of features is selected in terms of R^2 and MSE indexes. The performance evaluation indicators corresponding to different number of features are shown in Fig. 2.

The mean square error(MSE) is the average of the sum of the squares of the errors of the corresponding sample points of the experimental data and the fitted data. The smaller its value indicates a better fit. R^2 is the correlation coefficient between the original data and the fitted data. The closer the R^2 is to 1, the smaller the deviation between the experimental and fitted values [31]. As shown in Fig. 2, the model performance is optimal when the number of features is 10, and its

Table 4
Final parameter optimization results.

Parameter	Value range	Result
learning_rate	(0,1,0.001)	0.688
max_depth	(2,10)	4
n_estimators	(45,1000,5)	245
num_leaves	(2,60,2)	48
bagging_fraction	(0.3,1,0.1)	0.7
lambda_l1	(0.1,1,0.1)	0.3
lambda_l2	(0.1,1,0.1)	0.7

R^2 and MSE are 0.8613 and 13.3927, respectively. When the input features are larger than 10, the R^2 of the model gradually falls down, and the mean squared error tends to rise. This indicates that too many features cannot improve the performance of the model. Instead, it causes overfitting and reduces the generalization ability. Therefore, the top 10 features in the importance ranking, including average speed, average throttle pedal value, negative acceleration standard deviation, average positive acceleration, rapid acceleration ratio, average negative acceleration, throttle pedal standard deviation, long-duration idle ratio, positive acceleration standard deviation, and total mileage are selected as the set of the most optimal features to be input.

3.3. Parameter optimization

Parameter optimization is a crucial task in machine learning, which can greatly affect the performance of the algorithm. The traditional method of manual parameter tuning relies mainly on personal experience, which takes a long time and cannot guarantee the best combination of parameters. The same set of parameters exhibits different performance on different tasks. Optuna [32] is an automatic hyperparameter optimization framework designed for machine learning with a command-based, runtime-defined user API. Users are able to dynamically construct a hyperparameter search space based on actual needs, and then find the optimal hyperparameter combination results. Sklearn's GridSearchCV [33] function can be used for hyperparameter optimization by grid search and cross-validation though. It traverses all hyperparameter combinations within a given range and the debugging time is too long. Optuna internally uses the mechanism of Bayesian [34] tuning parameter to terminate the less probable parameter intervals by prediction, and use more arithmetic power on the more probable regions to improve the search efficiency. Compared with other methods, Optuna has the advantages of fast debugging speed, strong embedding and dynamic adjustment of parameter space. Therefore, Optuna is used in this paper for hyperparameter optimization of the LightGBM model.

The final tuned parameter optimization results are shown in Table 4. Compared with the R^2 (0.8613) and MSE (13.3927) of the model under the default parameters, the optimized R^2 and MSE are 0.8725 and 13.0416, respectively, which are increased by 1.3% and -2.62% . This indicates that Optuna significantly improves the prediction accuracy of LightGBM.

Table 5

Fuel consumption prediction results based on LightGBM model.

NO.	Real fuel consumption (L/100 km)	Predicted fuel consumption (L/100 km)	Absolute error (L/100 km)	Relative error	NO.	Real fuel consumption (L/100 km)	Predicted fuel consumption (L/100 km)	Absolute error (L/100 km)	Relative error
1	41.26	41.36	0.10	0.25%	21	30.10	29.89	-0.21	0.71%
2	43.85	44.62	0.77	1.75%	22	41.52	41.91	0.39	0.94%
3	42.83	41.95	-0.88	2.06%	23	47.43	47.30	-0.13	0.28%
4	25.27	24.56	-0.71	2.80%	24	41.30	38.68	-2.61	6.33%
5	42.67	40.53	-2.14	5.01%	25	38.81	40.35	1.54	3.96%
6	41.77	44.30	2.53	6.05%	26	58.14	59.53	1.40	2.40%
7	43.48	44.61	1.13	2.60%	27	36.67	37.30	0.63	1.72%
8	67.06	66.88	-0.18	0.27%	28	43.85	43.85	0.00	0.00%
9	56.72	56.11	-0.61	1.08%	29	37.66	36.17	-1.48	3.94%
10	56.33	56.23	-0.10	0.18%	30	27.50	29.60	2.10	7.64%
11	46.72	46.52	-0.19	0.42%	31	54.44	55.06	0.62	1.13%
12	43.50	42.77	-0.74	1.70%	32	39.24	39.10	-0.15	0.38%
13	47.48	46.74	-0.75	1.57%	33	40.51	43.54	3.04	7.49%
14	32.68	32.74	0.06	0.20%	34	43.81	43.51	-0.30	0.69%
15	26.13	26.90	0.77	2.94%	35	62.61	61.62	-0.99	1.58%
16	47.02	47.11	0.09	0.18%	36	27.70	28.03	0.33	1.17%
17	44.07	42.08	-2.00	4.53%	37	56.25	60.44	4.19	7.45%
18	42.09	44.50	2.42	5.75%	38	53.21	57.96	4.75	8.93%
19	36.07	36.78	0.71	1.98%	39	27.53	27.94	0.42	1.52%
20	44.17	46.75	2.58	5.84%	40	62.44	62.96	0.53	0.84%

4. Results and discussion

4.1. Predicted results

The LightGBM model prediction results after parameter optimization are compared with the real values in the corresponding test set. In these comparison groups, the smallest absolute error is 0 L/100 km and the largest absolute error is 4.75 L/100 km. The relative error ranges from 0 to 8.93%, with just under 20% of the samples having a relative error of more than 5%. From the statistical trend, the average predicted fuel consumption is 43.39 L/100 km, while the average actual fuel consumption is 43.67 L/100 km, with an average relative error of 0.69%. In addition, the standard deviation of the predicted values is 9.51 and the standard deviation of the true values is 10.55, which indicates that the LightGBM model is more constricted in its prediction and better fitted than the true trend. As shown in Table 5, 40 randomly selected prediction results are displayed to visualize the predictions of the model.

4.2. Model performance comparison

In order to comprehensively evaluate the performance of the LightGBM model constructed in this paper in segment fuel consumption prediction, common machine learning models such as XGboost, Random Forest [35], and Support Vector Machines [36] are selected for comparative analysis, respectively. Regression modeling is performed using the same training and test sets, and the best model after tuning and optimization is compared with LightGBM. The results of the comparative analysis are shown in Table 6. The data shows that XGBoost has the highest R^2 among all models, but it does not perform as well as LightGBM on MAE and MSE. This indicates that the XGBoost model predicted results are overfitted, which reduces the accuracy of its prediction. Comprehensively comparing the above several evaluation indexes, it can be concluded that LightGBM algorithm has the most excellent performance in segmented fuel consumption prediction.

4.3. Analysis of factors affecting fuel consumption of heavy duty trucks based on SHAP

Training based on the LightGBM algorithm can result in a prediction model with high prediction accuracy, but LightGBM performs poorly in terms of model interpretability compared to traditional statistical models. This paper introduces the interpretable machine learning

Table 6

Performance comparison of fragmented fuel consumption models.

Model	Evaluation indicators		
	MAE	MSE	R^2
LightGBM	2.0355	13.0416	0.8725
XGBoost	2.1385	13.5783	0.8937
RF	2.4062	14.3403	0.7684
SVM	2.3297	14.2481	0.8007

framework SHAP [37]. The SHAP values and feature visualization are used to visually express the effect of each feature on fuel consumption from different perspectives.

4.3.1. Global analysis

SHAP feature importance is the degree to which each feature contributes to improving the predictive power of the overall model [38]. According to Fig. 3, the SHAP model feature importance ranking shows that the feature with the highest contribution value to the model regression results is the average speed, followed by the average throttle pedal value, average positive acceleration, average negative acceleration, and so on. Due to the differences in the calculation method, the feature importance ranking of the SHAP output is not exactly the same as that of LightGBM. However, both of them put the average speed and the average throttle pedal value in the top two, which is a good indication that these two features have a strong influence on fuel consumption.

4.3.2. Single factor analysis

The single factor dependency graphs [39] are constructed to visualize each sample and study the specific impact of each feature on fuel consumption. As shown in Fig. 4(a), the average speed has a strong linear relationship with fuel consumption. In the range of (0, 80) km/h, the SHAP value turns from positive to negative as the average speed increases. This indicates that higher average speed can reduce the fuel consumption well. In reality, while pursuing fuel economy, it is necessary to ensure driving safety. Drivers should keep their speeds high and not excessive.

According to Fig. 4(b), it can be seen that the SHAP value turns from negative to positive as the average throttle pedal value increases while the average value of the gas pedal is within the range of (0, 30). When this interval is exceeded, there is a small decreasing fluctuation in the SHAP value before it continues to increase. This indicates that keeping the speed stable can reduce fuel consumption by reducing the frequent pressing of the throttle pedal.

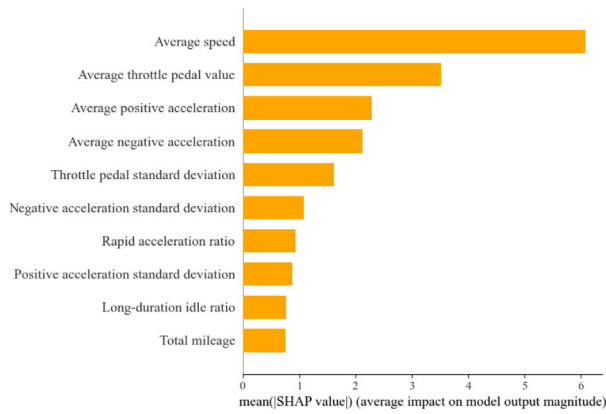


Fig. 3. SHAP importance histogram of each feature.

From Fig. 4(c), it can be observed that as the average negative acceleration increase, the SHAP value tends to be negative, indicating that this acts as a reduction in fuel consumption. The increase in average negative acceleration indicates that the vehicle speed tends to stabilize, and also indicates that the stabilization of the vehicle speed can effectively reduce fuel consumption. This is consistent with the above conclusion. The steady speed ensures lower fuel consumption while avoiding additional safety risks.

From Fig. 4(d), it can be found that most of the kinematic segments have long-duration idle in the 0%–10% range, implying a high proportion of uniform speed operating conditions. A higher long-duration idle ratio improves fuel consumption significantly, indicating the facilitating effect of smooth traffic flow on fuel economy. It also confirms the importance of observing the operating rules and turning off the engine in time [40].

Fig. 4(e) and (f) show that the negative and positive acceleration standard deviations follow a similar trend, both showing a “V” shape. At first, the SHAP value is positive in the case of lower standard deviation, indicating that the lower standard deviation corresponds to the increase of fuel consumption. When the positive or negative acceleration standard deviation starts to increase, the SHAP value starts to decrease to negative, which has a negative impact on fuel consumption. Then, with the further increase of the standard deviation, the SHAP value turns back to positive. According to this situation, this paper concludes that the acceleration standard deviation value should be reduced, with a negative acceleration standard deviation is about 1.0 and a positive acceleration standard deviation is about 0.6.

4.3.3. Dual factor interactions

Construct dual factor dependency graphs to explain the impact of dual factor interaction on SHAP. Select several sets of features with significant influence and close correlation for coupling [41]. As shown in Fig. 5(a), there is a close relationship between the average speed and the average throttle pedal value. When the average throttle pedal increases, it intuitively reflects the increase in the average vehicle speed and the decrease in SHAP value. This indicates that the driver can ensure better fuel conservation at high speeds by controlling the throttle pedal to a certain depth.

From Fig. 5(b), it can be seen that the lighter colored dots are concentrated in the case where the average negative acceleration is large, meaning that the average speed is higher. This corresponds to a negative SHAP value, implying lower fuel consumption. This indicates that keeping the vehicle in a high and stable speed range can greatly reduce fuel consumption. It also shows that smooth traffic and stable road conditions have a contributing effect on reducing fuel consumption.

From Fig. 5(c), it can be seen that in order to ensure timely transportation, when the driver is confronted with a more severe deceleration situation, the average positive acceleration corresponding to that

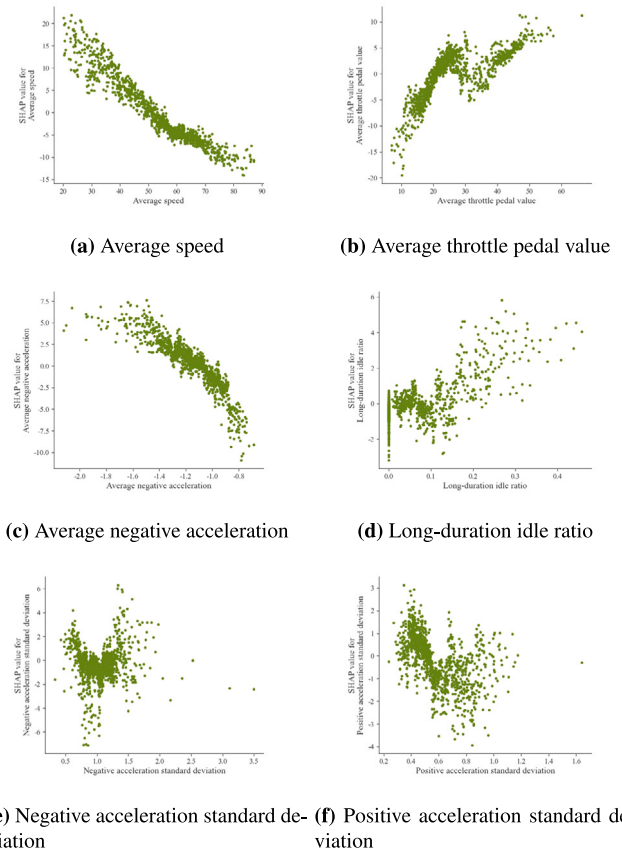


Fig. 4. Single factor analysis based on SHAP.

segment is accordingly boosted. This results in more rapid acceleration behavior and leads to more fuel consumption. As the average negative acceleration gradually decreases, the average positive acceleration of the segment also tends to decrease. At this point, the speed is in a more stable range, which has a significant effect on reducing fuel consumption.

From Fig. 5(d), it can be seen that when the average throttle pedal value is low, the standard deviation of the throttle pedal is concentrated at a lower position. The SHAP value generated at this point is positive, indicating an increase in fuel consumption in this segment. When the standard deviation of the throttle pedal is in the (20%,30%) range, the average value of the throttle pedal is concentrated in the (30%,40%) range. And the interaction of the two reduces fuel consumption. With a further increase in the standard deviation of the throttle pedal, the average value of the throttle pedal exceeds the range of 40%. Their interaction also leads to an increase in SHAP. Therefore, it can be concluded that maintaining the appropriate level of throttle pedal pressing and avoiding frequent control can reduce fuel consumption well.

As shown in Fig. 5(e), when the average speed of a kinematic segment is too low, its corresponding mileage is also at a lower level. This is due to poorer road conditions and traffic congestion forcing the vehicle to perform more long-duration idling and deceleration, eventually leading to stalling and stopping, which reduces fuel economy. And when the average speed is high, it can reflect that the segment has smooth roads and traffic. At this point, the vehicle exhibits a longer driving range while producing lower fuel consumption.

Fig. 5(f) shows how fuel consumption is affected by the interaction of rapid acceleration ratio and long-duration idle ratio. A large number of points in the figure are clustered in the upper left corner,

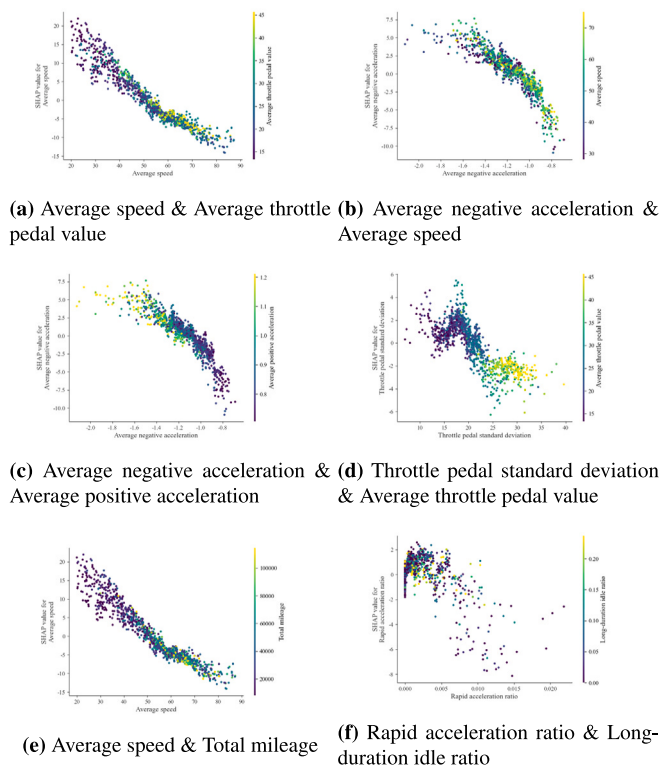


Fig. 5. Dual factor interactions analysis based on SHAP.

suggesting a lower rapid acceleration ratio in most of the segments. And one of the higher long-duration idle ratios leads to increased fuel consumption. When the rapid acceleration ratio increases, the color of the corresponding point is mostly darker, indicating a decrease in the long-duration idle ratio. This may be due to the fact that rapid acceleration avoids idling while waiting for traffic lights, resulting in a reduction in fuel consumption. However, the situation is unique and frequent rapid acceleration behavior is not advocated in consideration of driving safety.

5. Conclusions

In this paper, heavy duty trucks are used as research objects to generate kinematic segments based on a large amount of real-time vehicle operation data. The LightGBM algorithm is applied to segmented fuel consumption prediction, and the feature selection and parameter optimization of the prediction model are carried out to improve the prediction accuracy of the model. The results show that the optimized R^2 and MSE increased by 1.3% and -2.62% , respectively. Notably, our optimized LightGBM algorithm outperforms established algorithms such as XGBoost, Random Forest, and SVR in segmented fuel consumption prediction.

We use an interpretable method based on SHAP values to provide an intuitive explanation of the model prediction process. A feature importance ranking is established, and it is concluded that the factor with the greatest influence on segmented fuel consumption is average speed, followed by average throttle pedal value, average positive acceleration, and average negative acceleration. The single factor and dual factor interaction plots based on SHAP values show that fuel consumption can be effectively reduced by controlling the throttle pedal and ensuring that the vehicle speed is in a higher and more stable range. Reducing the frequency of acceleration and deceleration, and avoiding long-duration idling can also reduce fuel consumption. The relevant conclusions obtained using SHAP values and feature visualization are

highly compatible with the Operating specification for energy-saving driving of vehicle. The applicability of the optimized LightGBM algorithm and the interpretable method based on SHAP values for fuel consumption prediction of heavy duty trucks is verified. It provides an effective research methodology for the healthy development of the urban environment, the planning of staggered vehicle trips by road transportation enterprises and the evaluation of drivers' energy efficient driving techniques.

CRedit authorship contribution statement

Xinyi Zhu: Writing – original draft, Methodology, Conceptualization. **Xiaoyan Shen:** Writing – review & editing, Supervision. **Kailiang Chen:** Formal analysis, Data curation. **Zeqing Zhang:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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